Astrea Academy Trust
INSPIRING BEYOND MEASURE

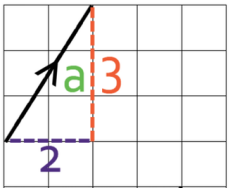
Name .....

Class .....

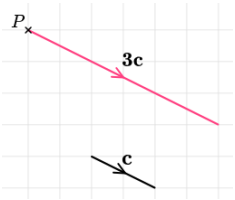
Tutor Group .....

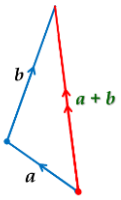
YEAR 11 CORE MATHEMATICS

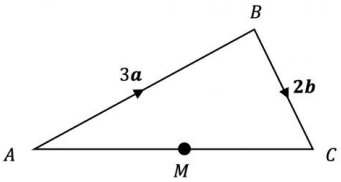
Booklet 3A – Vectors



$a = \binom{2}{3}$







KEY VOCABULARY LIST	
Direction	Where something is pointing.
Magnitude	The size of something. The magnitude of a number (also called its absolute value) is its distance from zero, so • the magnitude of 6 is 6 • the magnitude of −6 is also 6 The magnitude of a vector is its length (ignoring direction).
Vector	A vector quantity has both size and direction. Vectors can be added, subtracted and multiplied by a scalar. Geometrical problems can be solved using vectors.
Column Vector	Notation showing horizontal and vertical movement of a point: $\binom{x}{y}$
Resultant vector	A resultant vector is the sum of two or more vectors. It is the result of adding two or more vectors together.
Scalar	A number on its own (used when dealing with vectors or matrices). Example: we can multiply the vector (5,2) by the scalar 3 to get a new vector (15,6) 3 × (5,2) = (3×5,3×2) = (15,6)
Parallel	Two lines are parallel if they never meet, maintaining a constant distance between them.

CONTENTS		PAGE
SUBTOPIC 1	**COLUMN VECTORS**	**3**
SUBTOPIC 1A	Definition & Notation	3
SUBTOPIC 1B	Multiplying Vectors	7
SUBTOPIC 1C	Adding and Subtracting Vectors	10
SUBTOPIC 2	**PARALLEL VECTORS**	**16**
SUBTOPIC 3	**VECTOR GEOMETRY**	**19**
	EXAM STYLE QUESTIONS	**22**

BOOKLET STRUCTURE

Start of subtopic:

- Read and annotate the information in the Frayer model or introduction text
- I do (teacher led example).
- We do (teacher and scholar led example).
- You do (independent scholar example).
- Whiteboard check.
- Independent practice and mixed practice.

End of subtopic:

- Skills checks – key questions to help your teachers to pinpoint any issues.
- GCSE questions – to ensure you are confident with the style you will encounter in your exams.

Subtopic 1: Column Vectors

LEARNING OUTCOME: To understand and use column vector notation.

Objectives:
- Interpret vectors as a movement from one position to another.
- Use column notation to identify, describe and draw vectors on a grid.
- Use vector geometry to find pathways between points using vectors.

Keywords:
- Vector
- Scalar
- Magnitude
- Direction

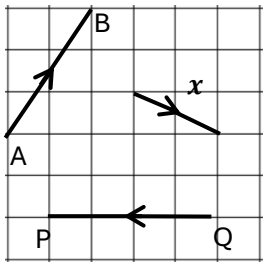
Vector

Definition:
A quantity with both direction and magnitude.

It describes the movement from one position to another.

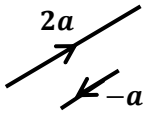
Characteristics:
Represented graphically by a line of a given length with an arrow to show direction.
Can be written using one letter notation, $\boldsymbol{a}$, two letter notation, $\overrightarrow{AB}$, or a column vector $\begin{pmatrix} x \\ y \end{pmatrix}$
Can be scaled, added and subtracted.

Examples:

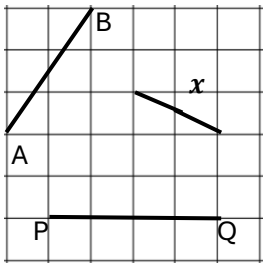


$\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\overrightarrow{AB}$

$\begin{pmatrix} -3 \\ 7 \end{pmatrix}$

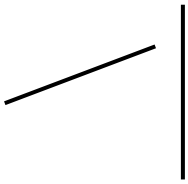


Non-examples:



$(2,3)$ AB

$\begin{pmatrix} \frac{2}{3} \end{pmatrix}$



Subtopic 1A: Definition & Notation

	Example 1. I do...	*We do...*	*You do...*
Horizontal Movement +: Right −: Left $\begin{pmatrix} x \\ y \end{pmatrix}$ $\begin{pmatrix} \leftrightarrow \\ \updownarrow \end{pmatrix}$ Vertical Movement +: Up −: Down	Write as a column vector: a) 2 right, 5 up b) 2 left, 5 up Write in words: $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$	Write as a column vector: a) 4 left, 3 up b) 10 right Write in words: $\begin{pmatrix} -6 \\ 1 \end{pmatrix}$	Write as a column vector: a) 7 left, 5 down b) 3 down Write in words: $\begin{pmatrix} -8 \\ 0 \end{pmatrix}$

Independent Practice

Question 1. Fill the gaps to give matching descriptions of the column vector & worded directions:

$a)$ $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$	6 _____ 3 _____	$b)$ $\begin{pmatrix} \square \\ 3 \end{pmatrix}$	6 left 3 _____
$c)$ $\begin{pmatrix} \square \\ \square \end{pmatrix}$	8 right 7 down	$d)$ $\begin{pmatrix} \square \\ 0 \end{pmatrix}$	3 left __ _____

Question 2. Write as a column vector:		**Question 3.** Write in words:	
a) 8 right 3 down	b) 9 left 2 up	$a)\ \begin{pmatrix}-2\\8\end{pmatrix}$	$b)\ \begin{pmatrix}3\\4\end{pmatrix}$
c) 6 down 7 right	d) 4 left	$c)\ \begin{pmatrix}0\\5\end{pmatrix}$	$d)\ \begin{pmatrix}-10\\-2\end{pmatrix}$

Question 4. Circle 'true' or 'false' for each of the following;

1) $\begin{pmatrix}4\\2\end{pmatrix}$ = →4↑2 True/False

2) $\begin{pmatrix}3\\2\end{pmatrix}$ = →3↑2 True/False

3) $\begin{pmatrix}-3\\2\end{pmatrix}$ = →3↑2 True/False

4) $\begin{pmatrix}-4\\2\end{pmatrix}$ = ←4↑2 True/False

5) $\begin{pmatrix}-1\\-1\end{pmatrix}$ = ←1↑1 True/False

6) $\begin{pmatrix}-2\\-2\end{pmatrix}$ = ←2↓2 True/False

7) $\begin{pmatrix}-3\\-3\end{pmatrix}$ = ←3↓3 True/False

8) $\begin{pmatrix}3\\-1\end{pmatrix}$ = →3↓1 True/False

9) $\begin{pmatrix}-3\\-1\end{pmatrix}$ = ←3↓1 True/False

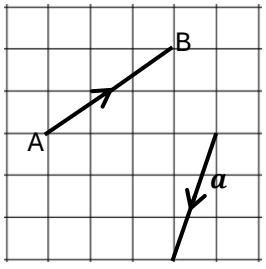
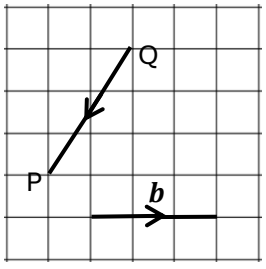
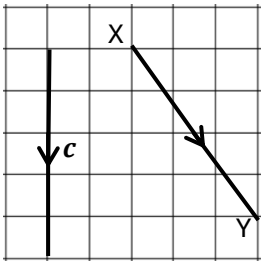
10) $\begin{pmatrix}4\\-1\end{pmatrix}$ = →4↑1 True/False

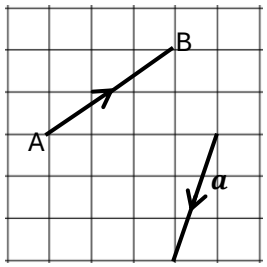
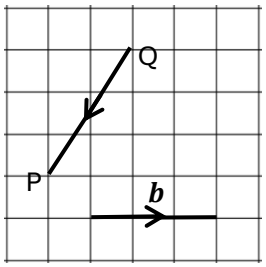
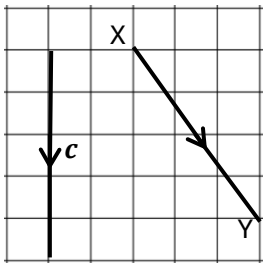
11) $\begin{pmatrix}5\\-1\end{pmatrix}$ = →5↓1 True/False

12) $\begin{pmatrix}6\\-1\end{pmatrix}$ = →6↑1 True/False

13) $\begin{pmatrix}7\\-1\end{pmatrix}$ = →7↓1 True/False

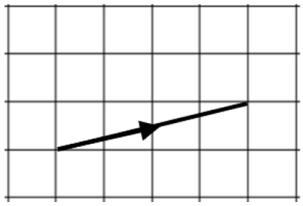
14) $\begin{pmatrix}-8\\0\end{pmatrix}$ = ←8↕0 True/False

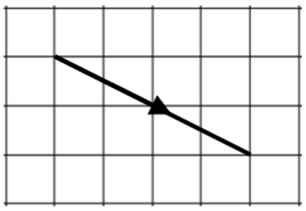
Steps	***Example 2. I do...***	***We do...***	***You do...***
1. Use the arrow to determine the start point of the vector. 2. Count the number of horizontal and vertical squares between the start and end points. 3. Write as a column vector.	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors. 

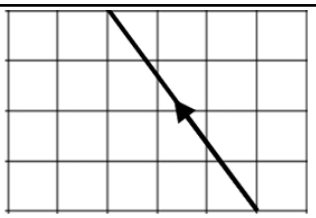
Steps	*Example 3. I do...*	*We do...*	*You do...*
1. Use the arrow to determine the start point of the vector. 2. Count the number of horizontal and vertical squares between the start and end points. 3. Write as a column vector.	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors.  	Write the vectors below as a column vectors. 

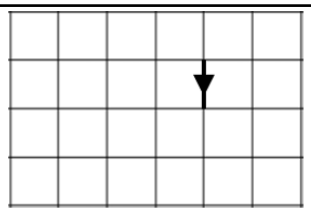
Independent Practice

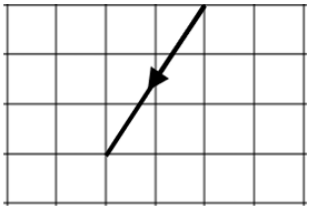
Question 5. Write as a column vector:

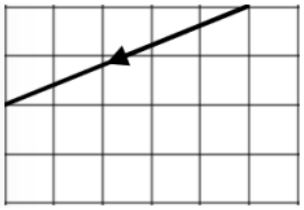
a) 

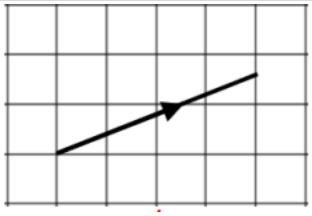
b) 

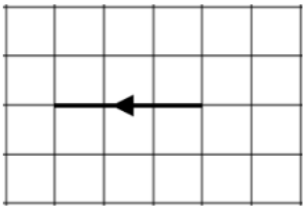
c) 

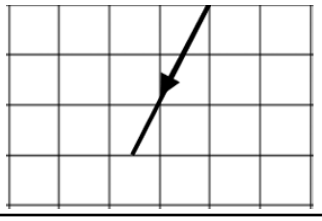
d) 

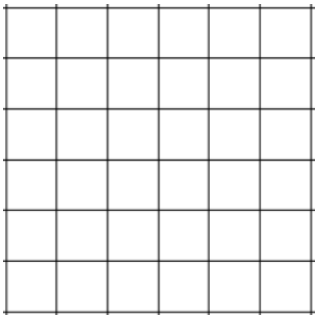
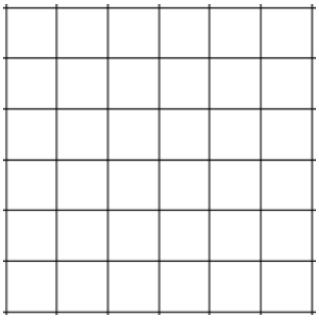
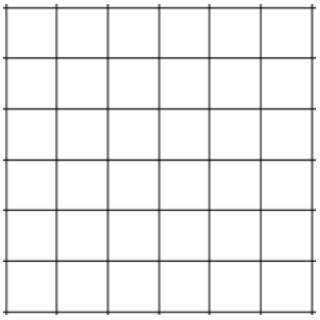
e) 

f) 

g) 

h) 

i) 

Steps	*Example 4. I do...*	*We do...*	*You do...*
1. Choose the start point of the vector. 2. Count the number of horizontal and vertical squares in the vector. 3. Draw in the end point. 4. Join the start and end points and draw the arrow to show the direction.	Draw the vectors: $\boldsymbol{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \boldsymbol{b} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ 	Draw the vectors: $\boldsymbol{c} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}, \boldsymbol{d} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$ 	Draw the vectors: $\boldsymbol{e} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}, \boldsymbol{f} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$ 

Question 6. Draw the following column vectors:

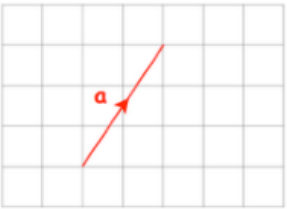
$\begin{pmatrix}5\\3\end{pmatrix}$ $\begin{pmatrix}0\\6\end{pmatrix}$ $\begin{pmatrix}2\\-3\end{pmatrix}$

$\begin{pmatrix}-4\\0\end{pmatrix}$ $\begin{pmatrix}-1\\-5\end{pmatrix}$ $\begin{pmatrix}0.5\\-2\end{pmatrix}$

Question 7.

Mark has been asked to draw the vector $\boldsymbol{a} = \begin{pmatrix}3\\2\end{pmatrix}$.

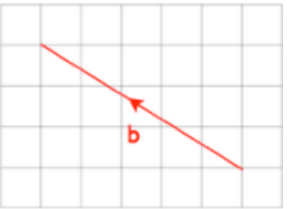
Explain his mistake.



Question 8.

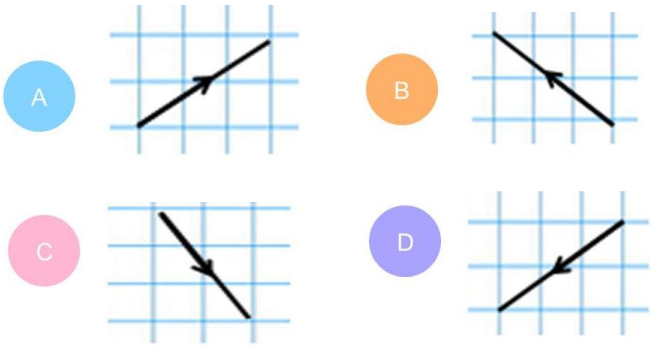
Abby has been asked to draw the vector $\boldsymbol{b} = \begin{pmatrix}5\\-3\end{pmatrix}$.

What mistake(s) has she made?

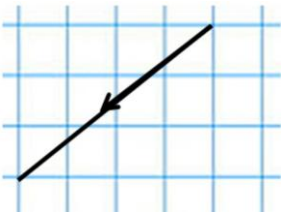


Skills Check

Which diagram shows the column vector $\begin{pmatrix}-3\\2\end{pmatrix}$?

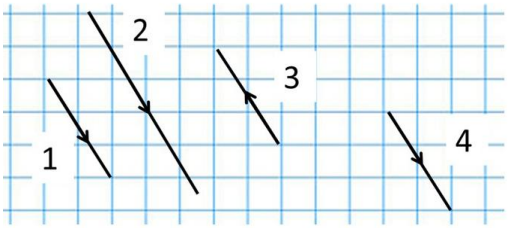


Which column vector matches the diagram?



A $\begin{pmatrix}4\\3\end{pmatrix}$ B $\begin{pmatrix}-3\\-4\end{pmatrix}$

C $\begin{pmatrix}4\\-3\end{pmatrix}$ D $\begin{pmatrix}-4\\-3\end{pmatrix}$



Which vectors in the diagram are the same?

A 1 and 4

B All

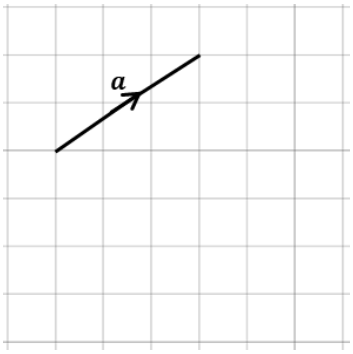
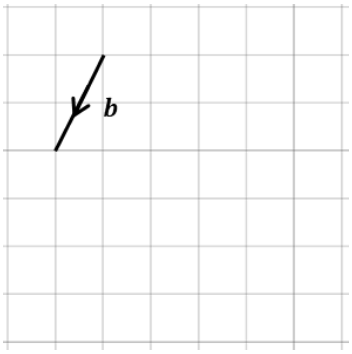
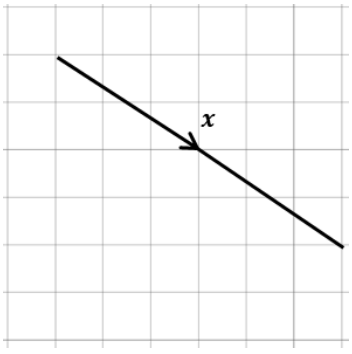
C 1, 3 and 4

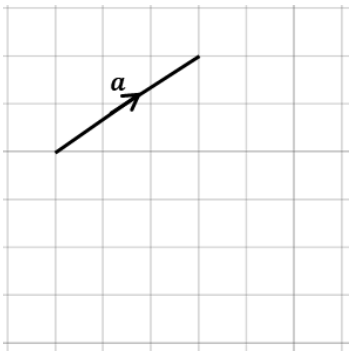
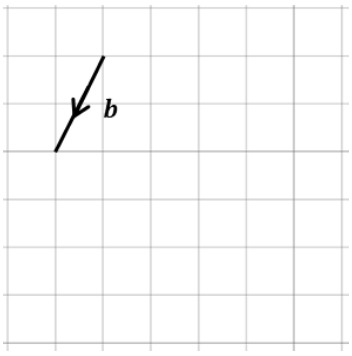
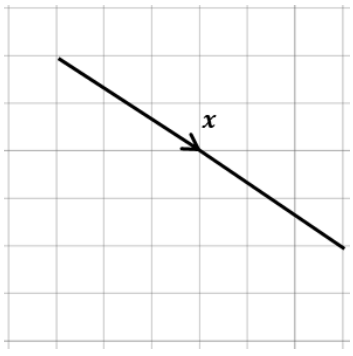
D None

Subtopic 1B: Multiplying Vectors

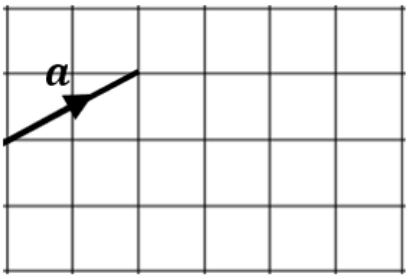
1. A **scalar** is a quantity with **only magnitude and no direction**. They are described by a **single**
2. **number**, and they often represent something measurable such as a length or a time
3. e.g. a measurement of 5m or a time of 24 seconds.
4. We can also **multiply vectors by a scalar**, to change the **magnitude;** make it longer or shorter.

5. If we multiply a vector by a **positive** scalar, the **direction** is **unchanged.**
6. If we multiply a vector by a **negative** scalar, the **direction** is **reversed**.

Steps	*Example 5. I do...*	*We do...*	*You do...*
1. Count the horizontal and vertical movement for the vector. 2. Pick a new start point and repeat this the required number of times.	The vector $\boldsymbol{a}$ is shown below. Draw the vector $2\boldsymbol{a}$. 	The vector $\boldsymbol{b}$ is shown below. Draw the vector $3\boldsymbol{b}$. 	The vector $\boldsymbol{x}$ is shown below. Draw the vector $\frac{1}{2}\boldsymbol{x}$. 
	Write as column vectors:	Write as column vectors:	Write as column vectors:

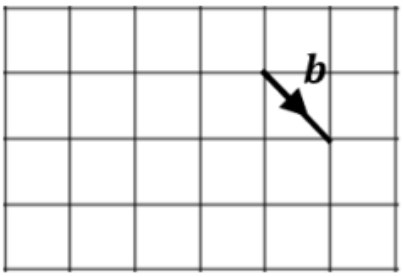
Steps	*Example 6. I do...*	*We do...*	*You do...*
1. Count the horizontal and vertical movement for the vector. 2. Pick a new start point and repeat this the required number of times in the opposite direction.	The vector $\boldsymbol{a}$ is shown below. Draw the vector $-\boldsymbol{a}$. 	The vector $\boldsymbol{b}$ is shown below. Draw the vector $-3\boldsymbol{b}$. 	The vector $\boldsymbol{x}$ is shown below. Draw the vector $-\frac{1}{2}\boldsymbol{x}$. 
	Write as column vectors:	Write as column vectors:	Write as column vectors:

Q9. Vector $\boldsymbol{a}$ is shown.
Draw vector $2\boldsymbol{a}$.



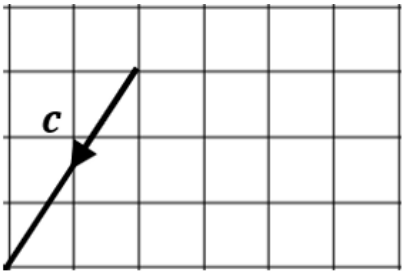
Write as column vectors:
$\boldsymbol{a} =$ $\qquad$ $2\boldsymbol{a} =$

Q10. Vector $\boldsymbol{b}$ is shown.
Draw vector $3\boldsymbol{b}$.



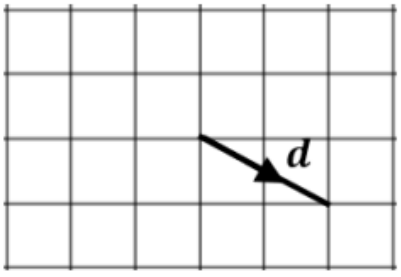
Write as column vectors:
$\boldsymbol{b} =$ $\qquad$ $3\boldsymbol{b} =$

Q11. Vector $\boldsymbol{c}$ is shown.
Draw vector $-\boldsymbol{c}$.



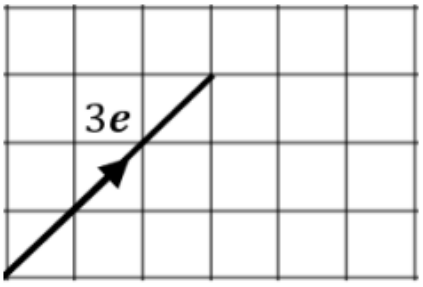
Write as column vectors:
$\boldsymbol{c} =$ $\qquad$ $-\boldsymbol{c} =$

Q12. Vector $\boldsymbol{d}$ is shown.
Draw vector $-2\boldsymbol{d}$.



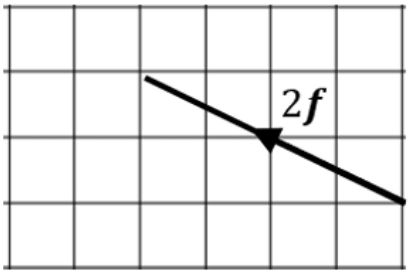
Write as column vectors:
$\boldsymbol{d} =$ $\qquad$ $-2\boldsymbol{d} =$

Q13. Vector $3\boldsymbol{e}$ is shown.
Draw vector $2\boldsymbol{e}$.



Write as column vectors:
$3\boldsymbol{e} =$ $\qquad$ $2\boldsymbol{e} =$

Q14. Vector $2\boldsymbol{f}$ is shown.
Draw vector $-\boldsymbol{f}$.



Write as column vectors:
$2\boldsymbol{f} =$ $\qquad$ $-\boldsymbol{f} =$

Steps	*Example 7. I do...*	*We do...*	*You do...*
1. Multiply both the horizontal and vertical components by the coefficient.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$ Find: 1) $4\boldsymbol{a}$ 2) $-\boldsymbol{a}$ 3) $\frac{1}{2}\boldsymbol{a}$	$\boldsymbol{c} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$ Find: 1) $5\boldsymbol{c}$ 2) $-2\boldsymbol{c}$ 3) $\frac{1}{10}\boldsymbol{c}$	$\boldsymbol{p} = \begin{pmatrix} -8 \\ -20 \end{pmatrix}$ Find: 1) $0.5\boldsymbol{p}$ 2) $-3\boldsymbol{p}$ 3) $-\frac{1}{4}\boldsymbol{p}$

Q15. John writes: $2\begin{pmatrix}-3\\7\end{pmatrix}=\begin{pmatrix}-6\\7\end{pmatrix}$

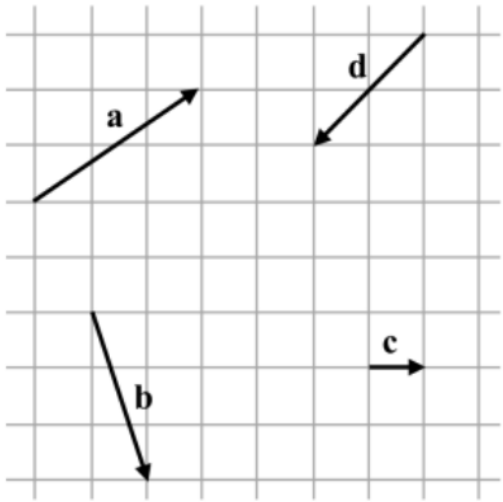
He is incorrect.
Explain why & give the correct answer.

Q16. Given that: $\boldsymbol{a}=\begin{pmatrix}2\\11\end{pmatrix}$ $\mathbf{b}=\begin{pmatrix}-8\\3\end{pmatrix}$ and $\boldsymbol{c}=\begin{pmatrix}-4\\-6\end{pmatrix}$

Write the following as column vectors.

a) $3\boldsymbol{a}$	b) $\frac{1}{2}\boldsymbol{c}$	c) $-\boldsymbol{a}$	d) $-\boldsymbol{b}$
e) $-\boldsymbol{c}$	f) $-2\boldsymbol{a}$	g) $-\frac{1}{2}\boldsymbol{c}$	h) $-5\mathbf{b}$

Q17. Work out the following, giving your answers as column vectors:



a) $2\boldsymbol{c}$

b) $2\boldsymbol{a}$

c) $3\boldsymbol{b}$

d) $\frac{1}{2}d$

e) $-\boldsymbol{c}$

f) $-\boldsymbol{b}$

g) $-5\boldsymbol{a}$

h) $-\frac{1}{2}\boldsymbol{d}$

Skills Check

$\boldsymbol{p}=\begin{pmatrix}-4\\8\end{pmatrix}$

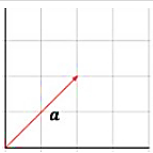
$3\boldsymbol{p}=?$

A $\begin{pmatrix}-1\\11\end{pmatrix}$ B $\begin{pmatrix}-12\\8\end{pmatrix}$ C $\begin{pmatrix}-12\\24\end{pmatrix}$ D $\begin{pmatrix}12\\24\end{pmatrix}$

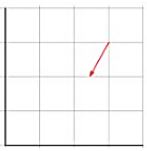
$-5\begin{pmatrix}3\\-4\end{pmatrix}=$

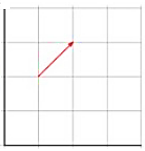
A $\begin{pmatrix}-15\\-20\end{pmatrix}$ B $\begin{pmatrix}-15\\20\end{pmatrix}$ C $\begin{pmatrix}-15\\-4\end{pmatrix}$ D $\begin{pmatrix}-2\\-9\end{pmatrix}$

The grid shows vector $\boldsymbol{a}$.
Which of the following is $\frac{1}{2}\boldsymbol{a}$?



A

B 

C 

D

$\boldsymbol{p}=\begin{pmatrix}3\\-6\end{pmatrix}$

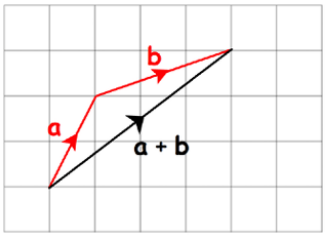
$-\frac{2}{3}\boldsymbol{p}=?$

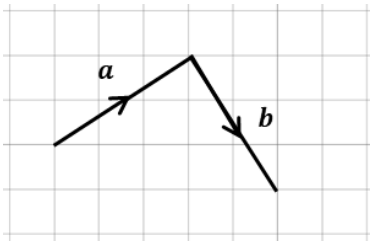
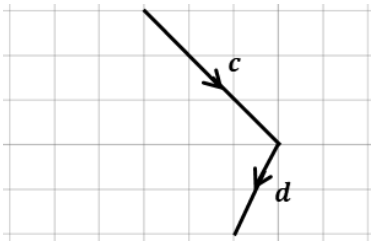
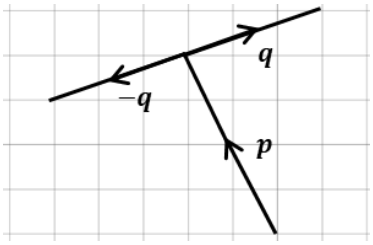
A $\begin{pmatrix}1\\-9\end{pmatrix}$ B $\begin{pmatrix}2\\-4\end{pmatrix}$ C $\begin{pmatrix}-2\\4\end{pmatrix}$ D $\begin{pmatrix}-1\\2\end{pmatrix}$

Subtopic 1C: Adding and Subtracting Vectors

1. We can add vectors together to create a combined movement, like
2. joining steps to form a path.
3. This new path is called the **resultant vector** – it replaces all the joined
4. vectors with a single one.

5. To add vectors, we combine their horizontal and vertical components
6. separately.

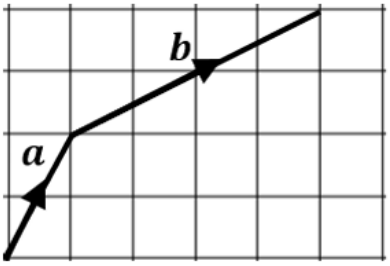


Steps	*Example 8. I do...*	*We do...*	*You do...*
1. Find the start point of the first vector. 2. Follow the given pathway to the end point of the second vector. 3. Join these 2 points and draw an arrow in the correct direction.	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $\boldsymbol{a}+\boldsymbol{b}$	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $\boldsymbol{c}+\boldsymbol{d}$	Vectors $\boldsymbol{p}$ & $\boldsymbol{q}$ are shown below.  Draw the vector $\boldsymbol{p}+\boldsymbol{q}$
	Write this addition in column vector notation.	Write this addition in column vector notation.	Write this addition in column vector notation.

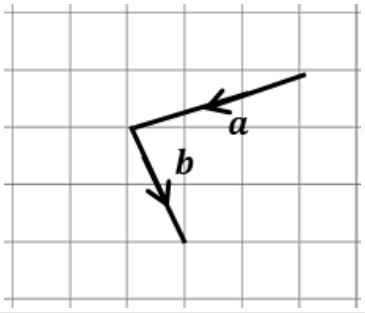
Steps	*Example 9. I do...*	*We do...*	*You do...*
1. Write the addition using column vectors. 2. Add the horizontal parts. 3. Add the vertical parts.	$\boldsymbol{a}=\begin{pmatrix}6\\1\end{pmatrix}$, $\boldsymbol{b}=\begin{pmatrix}4\\3\end{pmatrix}$ Calculate $\boldsymbol{a}+\boldsymbol{b}$	$\boldsymbol{x}=\begin{pmatrix}-3\\5\end{pmatrix}, \boldsymbol{y}=\begin{pmatrix}2\\1\end{pmatrix}$ Calculate $\boldsymbol{x}+\boldsymbol{y}$	$\boldsymbol{m}=\begin{pmatrix}5\\-1\end{pmatrix}, \boldsymbol{n}=\begin{pmatrix}-7\\-3\end{pmatrix}$ Calculate $\boldsymbol{m}+\boldsymbol{n}$

Independent Practice

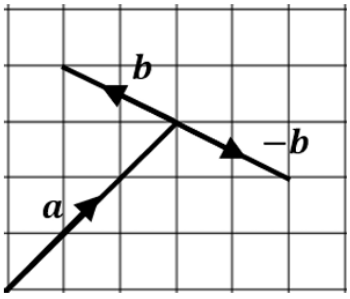
Q18. Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown. Draw vector $\boldsymbol{a}+\boldsymbol{b}$.



Q19. Vectors $\boldsymbol{a}$, & $\boldsymbol{b}$ are shown. Draw $\boldsymbol{b}+\boldsymbol{a}$.

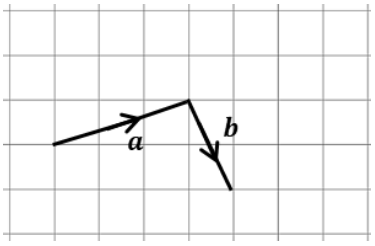
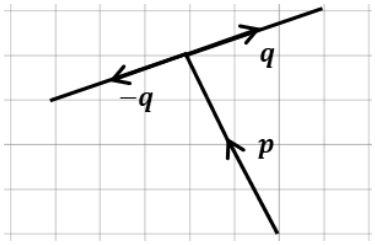
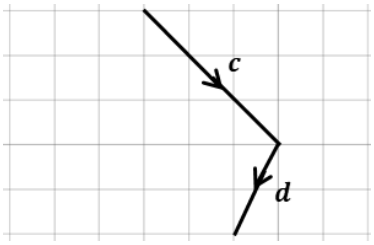


Q20. Vectors $\boldsymbol{a}$, $\boldsymbol{b}$ & $-\boldsymbol{b}$ are shown. Draw $\mathbf{a}+\boldsymbol{b}$.



Q21.	Q22.	Q23.
$\boldsymbol{x} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$	$\boldsymbol{a} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$	$\boldsymbol{m} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad \boldsymbol{n} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$
Calculate $\boldsymbol{x} + \boldsymbol{y}$	Calculate $\boldsymbol{a} + \boldsymbol{b}$	Calculate $\boldsymbol{m} + \boldsymbol{n}$

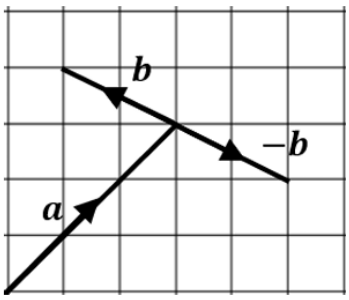
We treat **subtraction** of a vector as adding the vector in the **reverse direction**.

Steps	*Example 10. I do...*	*We do...*	*You do...*
1. Draw in the negative (reverse) vector where required. 2. Follow the pathway as before. 3. Join the start and end points, with an arrow in the correct direction.	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $\boldsymbol{a} - \boldsymbol{b}$	Vectors $\boldsymbol{p}$, $\boldsymbol{q}$ & $-\boldsymbol{q}$ are shown below.  Draw the vector $\boldsymbol{p} - \boldsymbol{q}$	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $\boldsymbol{d} - \boldsymbol{c}$
	Write this in column vector notation.	Write this in column vector notation.	Write this in column vector notation.

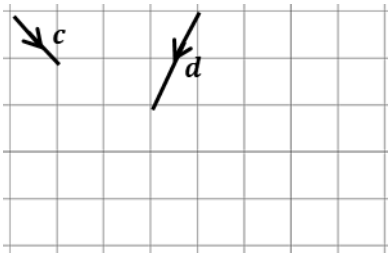
Steps	*Example 11. I do...*	*We do...*	*You do...*
1. Write the addition using column vectors. 2. Subtract the horizontal parts. 3. Subtract the vertical parts.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Calculate $\boldsymbol{a} - \boldsymbol{b}$	$\boldsymbol{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \boldsymbol{y} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ Calculate $\boldsymbol{y} - \boldsymbol{x}$	$\boldsymbol{m} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \boldsymbol{n} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$ Calculate $\boldsymbol{m} - \boldsymbol{n}$

Independent Practice

Q24. Vectors $\boldsymbol{a}$, $\boldsymbol{b}$ & $-\boldsymbol{b}$ are shown.
Draw $\mathbf{a} - \boldsymbol{b}$

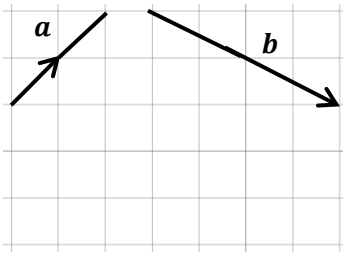
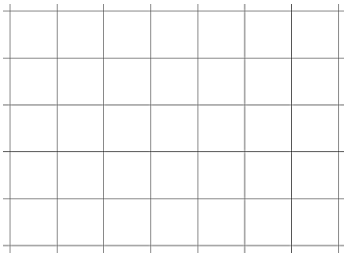
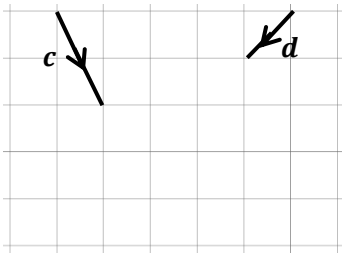
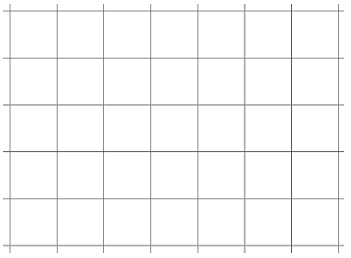
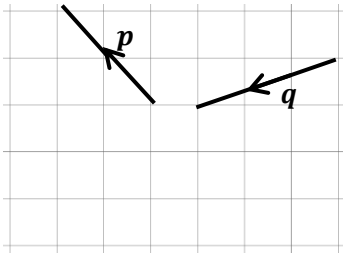
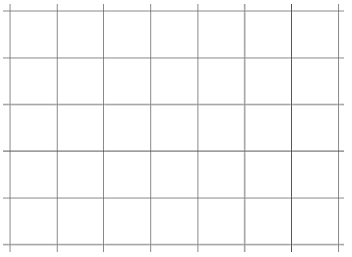


Q25. Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown. Draw $\boldsymbol{c} - \boldsymbol{d}$.



Q26.	Q27.	Q28.
$\boldsymbol{x} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \quad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ Calculate $\boldsymbol{x} - \boldsymbol{y}$	$\boldsymbol{a} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ Calculate $\boldsymbol{a} - \boldsymbol{b}$	$\boldsymbol{m} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad \boldsymbol{n} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$ Calculate $\boldsymbol{m} - \boldsymbol{n}$

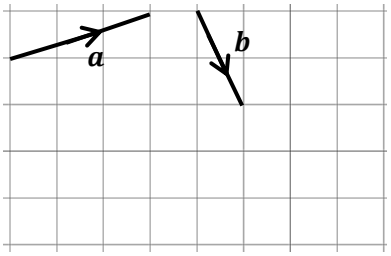
We can also add and subtract **multiples of vectors** to give resultant vectors.

Steps	*Example 12. I do...*	*We do...*	*You do...*
1. Write the column form for the vectors shown. 2. Calculate any multiples in column notation. 3. Draw the relevant single vectors to create a pathway.	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $2\boldsymbol{a} + \boldsymbol{b}$ 	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $\boldsymbol{c} + 3\boldsymbol{d}$ 	Vectors $\boldsymbol{p}$ & $\boldsymbol{q}$ are shown below.  Draw the vector $\frac{1}{2}\boldsymbol{p} + \boldsymbol{q}$ 
	Write this addition in column vector notation.	Write this addition in column vector notation.	Write this addition in column vector notation.

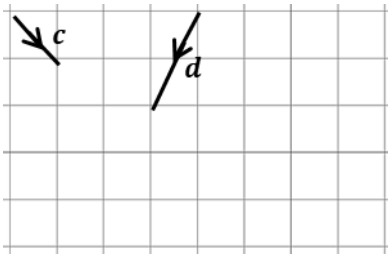
Steps	*Example 13. I do...*	*We do...*	*You do...*
1. Calculate any multiples required in column notation. 2. Complete the calculation using the horizontal and vertical parts separately.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Calculate $2\boldsymbol{a} + \boldsymbol{b}$	$\boldsymbol{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \boldsymbol{y} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ Calculate $3\boldsymbol{y} + 2\boldsymbol{x}$	$\boldsymbol{m} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \boldsymbol{n} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$ Calculate $4\boldsymbol{m} + 3\boldsymbol{n}$

Independent Practice

Q29. Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown.
Draw $\mathbf{a} + 2\boldsymbol{b}$



Q30. Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown. Draw $2\boldsymbol{c} + \boldsymbol{d}$.



Q31.

$$\boldsymbol{x} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}, \qquad \boldsymbol{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Calculate $\boldsymbol{x} + 4\boldsymbol{y}$

Q32.

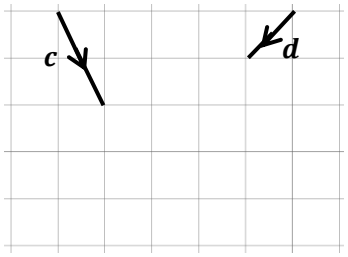
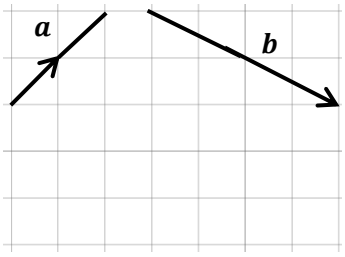
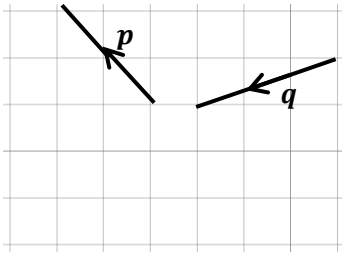
$$\boldsymbol{a} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

Calculate $\mathbf{3}\boldsymbol{a} + \mathbf{2}\boldsymbol{b}$

Q33.

$$\boldsymbol{a} = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$$

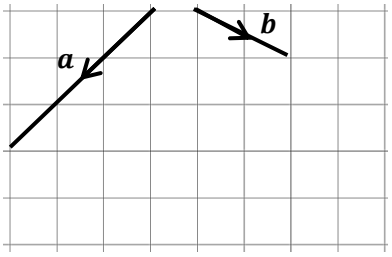
Calculate $\mathbf{5}\boldsymbol{a} + \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{b}$

Steps	*Example 14. I do...*	*We do...*	*You do...*
1. Write the column form for the vectors shown. 2. Calculate any multiples in column notation. (Remember that subtraction can be written as the addition of the reverse vector.) 3. Draw the relevant single vectors to create a pathway.	Vectors $\boldsymbol{c}$ & $\boldsymbol{d}$ are shown below.  Draw the vector $2\boldsymbol{c} - \boldsymbol{d}$	Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown below.  Draw the vector $\boldsymbol{a} - \frac{1}{2}\boldsymbol{b}$	Vectors $\boldsymbol{p}$ & $\boldsymbol{q}$ are shown below.  Draw the vector $-\mathbf{2}\boldsymbol{q} - \boldsymbol{p}$
	Write this addition in column vector notation.	Write this addition in column vector notation.	Write this addition in column vector notation.

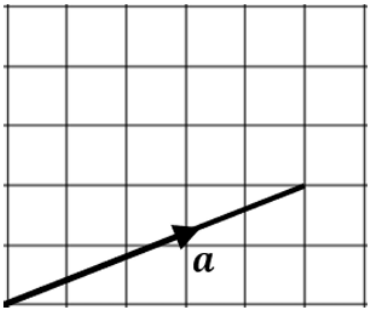
Steps	Example 15. I do...	We do...	You do...
1. Calculate any multiples required in column notation. 2. Complete the calculation using the horizontal and vertical parts separately.	$\boldsymbol{a} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$, $\boldsymbol{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ Calculate $2\boldsymbol{a} - \boldsymbol{b}$	$\boldsymbol{x} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \boldsymbol{y} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ Calculate $3\boldsymbol{y} - \frac{1}{2}\boldsymbol{x}$	$\boldsymbol{m} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \boldsymbol{n} = \begin{pmatrix} -7 \\ -3 \end{pmatrix}$ Calculate $4\boldsymbol{m} - 3\boldsymbol{n}$

Mixed Practice

Q34. Vectors $\boldsymbol{a}$ & $\boldsymbol{b}$ are shown.
Draw vector $\frac{1}{3}\mathbf{a} + 2\boldsymbol{b}$.



Q35.
$\boldsymbol{a} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$
$\boldsymbol{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$
Draw the vector $\boldsymbol{a} - \boldsymbol{b}$ and find its column vector.



Q36. $\begin{pmatrix} 3 \\ -2 \end{pmatrix} + 2\begin{pmatrix} -7 \\ 3 \end{pmatrix} =$

Q37. $3\begin{pmatrix} 5 \\ -6 \end{pmatrix} - \begin{pmatrix} -8 \\ 2 \end{pmatrix} =$

Q38. $4\begin{pmatrix} -5 \\ 3 \end{pmatrix} - 3\begin{pmatrix} -3 \\ -8 \end{pmatrix} =$

Q39. Given that $\boldsymbol{a} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$ $\boldsymbol{c} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$ and $\boldsymbol{d} = \begin{pmatrix} -4 \\ -3 \end{pmatrix}$, calculate:

a) $\boldsymbol{a} + \boldsymbol{c}$	b) $\boldsymbol{b} - \boldsymbol{d}$	c) $\boldsymbol{a} + \boldsymbol{c} - \boldsymbol{b}$	d) $2\boldsymbol{a} + 0.5\boldsymbol{c}$
e) $3\boldsymbol{c} - \boldsymbol{a}$	f) $-3\boldsymbol{d} + 5\boldsymbol{b}$	g) $3(\boldsymbol{a} - \boldsymbol{b})$	h) $2\boldsymbol{a} - \boldsymbol{b} + \frac{1}{4}\boldsymbol{c} - \boldsymbol{d}$

Q40. Find the values of x and y if:

$$\begin{pmatrix}14\\-4\end{pmatrix}+\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}10\\3\end{pmatrix}$$

Q41. Find the values of p & q if:

$$\begin{pmatrix}4\\5\end{pmatrix}-2\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}-6\\7\end{pmatrix}$$

Q42. Find the values of g and h if:

$$g\begin{pmatrix}5\\4\end{pmatrix}-3\begin{pmatrix}2\\6\end{pmatrix}=\begin{pmatrix}9\\h\end{pmatrix}$$

Q43. Find the values of p & q if:

$$\begin{pmatrix}-3\\7\end{pmatrix}-p\begin{pmatrix}2\\-4\end{pmatrix}=\begin{pmatrix}-13\\q\end{pmatrix}$$

Skills Check

$$\begin{pmatrix}\mathbf{5}\\\mathbf{2}\end{pmatrix}+\begin{pmatrix}\mathbf{6}\\\mathbf{-7}\end{pmatrix}=\ ?$$

A $\begin{pmatrix}30\\-14\end{pmatrix}$ B $\begin{pmatrix}12\\-35\end{pmatrix}$

C $\begin{pmatrix}11\\-5\end{pmatrix}$ D $\begin{pmatrix}11\\9\end{pmatrix}$

$$\begin{pmatrix}\mathbf{5}\\\mathbf{2}\end{pmatrix}-\begin{pmatrix}\mathbf{6}\\\mathbf{-7}\end{pmatrix}=\ ?$$

A $\begin{pmatrix}-1\\-5\end{pmatrix}$ B $\begin{pmatrix}-1\\5\end{pmatrix}$

C $\begin{pmatrix}11\\-5\end{pmatrix}$ D $\begin{pmatrix}-1\\9\end{pmatrix}$

$$\begin{pmatrix}\mathbf{5}\\\mathbf{2}\end{pmatrix}+\ ?\ =\begin{pmatrix}\mathbf{6}\\\mathbf{-7}\end{pmatrix}$$

A $\begin{pmatrix}1\\-5\end{pmatrix}$ B $\begin{pmatrix}1\\-9\end{pmatrix}$

C $\begin{pmatrix}11\\-5\end{pmatrix}$ D $\begin{pmatrix}-1\\9\end{pmatrix}$

c = $\begin{pmatrix}-2\\-5\end{pmatrix}$ **d** = $\begin{pmatrix}4\\-4\end{pmatrix}$ Work out the vector 2**c** + 3**d**

A $\begin{pmatrix}8\\2\end{pmatrix}$ B $\begin{pmatrix}8\\-22\end{pmatrix}$

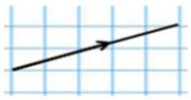
C $\begin{pmatrix}-48\\-22\end{pmatrix}$ D $\begin{pmatrix}8\\120\end{pmatrix}$

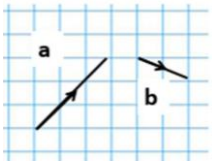
e = $\begin{pmatrix}-4\\9\end{pmatrix}$ **f** = $\begin{pmatrix}-3\\4\end{pmatrix}$ Work out the vector **e** – 2**f**

A $\begin{pmatrix}10\\-1\end{pmatrix}$ B $\begin{pmatrix}-10\\1\end{pmatrix}$

C $\begin{pmatrix}2\\1\end{pmatrix}$ D $\begin{pmatrix}2\\-1\end{pmatrix}$

The grid shows vectors $\boldsymbol{a}$ & $\boldsymbol{b}$. Which of the following is shown?

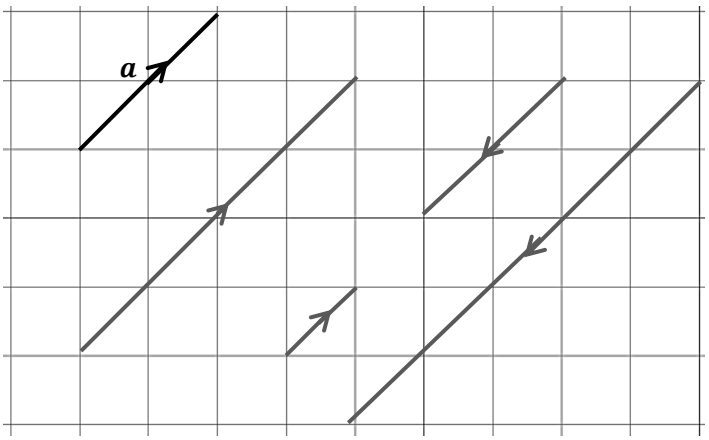




A $\boldsymbol{a}+\boldsymbol{b}$ B $\boldsymbol{a}-\boldsymbol{b}$

C $\boldsymbol{a}+2\boldsymbol{b}$ D None of these

Subtopic 2: Parallel Vectors



What is the same?

What is different?

Write these vectors in terms of $\boldsymbol{a}$

Write each vector in column notation.

Vectors are parallel if _____.

Steps	*Example 1. I do...*	*We do...*	*You do...*
1) Find a multiple, K, linking the two vectors such that $K\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}x\\y\end{pmatrix}$ 2) Write this as $\mathrm{K}\boldsymbol{a}=\boldsymbol{x}$	$\boldsymbol{a}=\begin{pmatrix}2\\-3\end{pmatrix}$ Show that the vector $\begin{pmatrix}6\\-9\end{pmatrix}$ is parallel to $\boldsymbol{a}$.	$\boldsymbol{b}=\begin{pmatrix}12\\-20\end{pmatrix}$ Show that $\begin{pmatrix}-3\\5\end{pmatrix}$ is parallel to $\boldsymbol{b}$.	$\boldsymbol{c}=\begin{pmatrix}-6\\10\end{pmatrix}$ Show that the vector $\begin{pmatrix}-15\\25\end{pmatrix}$ is parallel to $\boldsymbol{c}$.

Steps	*Example 2. I do...*	*We do...*	*You do...*
1) Find a multiple, K, linking the two vectors from the given components. 2) Write this as $\mathrm{K}\boldsymbol{a}=\boldsymbol{x}$ 3) Find the missing value	$\boldsymbol{a}=\begin{pmatrix}-7\\3\end{pmatrix}$ If $\begin{pmatrix}-21\\x\end{pmatrix}$ is parallel to $\boldsymbol{a}$, find the value of x.	$\boldsymbol{b}=\begin{pmatrix}24\\18\end{pmatrix}$ $\begin{pmatrix}-4\\y\end{pmatrix}$ is parallel to $\boldsymbol{b}$. Find the value of y.	$\boldsymbol{c}=\begin{pmatrix}-12\\21\end{pmatrix}$ $\boldsymbol{c}$ is parallel to $\begin{pmatrix}30\\z\end{pmatrix}$ Find the value of z.

Steps	Example 3. I do...	We do...	You do...
1) Complete the required column vector arithmetic 2) Find a multiple, K, linking the resultant and given vector. 3) Write this as $\mathrm{K}\boldsymbol{a} = \boldsymbol{x}$	$\boldsymbol{a} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ Show that $\boldsymbol{a} + \boldsymbol{b}$ is parallel to the vector $\begin{pmatrix} 15 \\ 3 \end{pmatrix}$.	$\boldsymbol{a} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ Show that $\boldsymbol{a} - \boldsymbol{b}$ is parallel to the vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.	$\boldsymbol{a} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$ Show that $2\boldsymbol{a} + \boldsymbol{b}$ is parallel to the vector $\begin{pmatrix} -15 \\ -9 \end{pmatrix}$

Steps	Example 4. I do...	We do...	You do...
1) Complete the required column vector arithmetic 2) Find a multiple, K, linking the resultant and given vector components. 3) Write this as $\mathrm{K}\boldsymbol{a} = \boldsymbol{x}$ 4) Find the missing value	$\boldsymbol{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ If $\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\begin{pmatrix} x \\ 12 \end{pmatrix}$, find the value of x.	$\boldsymbol{a} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -6 \\ 7 \end{pmatrix}$ If $\boldsymbol{a} - \boldsymbol{b}$ is parallel to $\begin{pmatrix} -5 \\ y \end{pmatrix}$, find the value of y.	$\boldsymbol{a} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$ $\boldsymbol{b} = \begin{pmatrix} -8 \\ -2 \end{pmatrix}$ If $2\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\begin{pmatrix} z \\ -9 \end{pmatrix}$, find the value of z.

Independent Practice

Question 1. Complete the table:

Vector 1	Vector 2	Parallel?	Same direction?	Same length?
$2\boldsymbol{b}$	$5\boldsymbol{b}$			
$2\boldsymbol{a}$	$-\boldsymbol{a}$			
$3\boldsymbol{p}$		Yes	No	Yes
$\begin{pmatrix} -1 \\ 5 \end{pmatrix}$	$\begin{pmatrix} -4 \\ \boxed{} \end{pmatrix}$	Yes		
$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$	$\begin{pmatrix} 9 \\ 16 \end{pmatrix}$			
$\begin{pmatrix} 15 \\ -30 \end{pmatrix}$	$\begin{pmatrix} -2 \\ 4 \end{pmatrix}$			
$\begin{pmatrix} -3 \\ -8 \end{pmatrix}$	$\begin{pmatrix} \boxed{} \\ \boxed{} \end{pmatrix}$	Yes	Yes	Yes

Question 2. Circle the parallel vectors in each set:

a) $\begin{pmatrix}20\\8\end{pmatrix}$ $\begin{pmatrix}5\\2\end{pmatrix}$ $\begin{pmatrix}-2\\-5\end{pmatrix}$ $\begin{pmatrix}-10\\4\end{pmatrix}$

b) $\begin{pmatrix}-2\\7\end{pmatrix}$ $\begin{pmatrix}4\\5\end{pmatrix}$ $\begin{pmatrix}-12\\15\end{pmatrix}$ $\begin{pmatrix}-16\\-20\end{pmatrix}$

c) $\begin{pmatrix}15\\-20\end{pmatrix}$ $\begin{pmatrix}14\\-21\end{pmatrix}$ $\begin{pmatrix}-10\\15\end{pmatrix}$ $\begin{pmatrix}4\\-9\end{pmatrix}$

Question 3.

$$\boldsymbol{b} = \begin{pmatrix}4\\3\end{pmatrix}$$

Show that $\begin{pmatrix}36\\27\end{pmatrix}$ is parallel to $\boldsymbol{b}$.

Question 4.

$$\boldsymbol{a} = \begin{pmatrix}4\\-6\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-5\\0\end{pmatrix}$$

Show that $\boldsymbol{a} - 2\boldsymbol{b}$ is parallel to $\begin{pmatrix}-21\\9\end{pmatrix}$.

Question 5.

$$\boldsymbol{a} = \begin{pmatrix}x\\4\end{pmatrix}$$

$\boldsymbol{a}$ is parallel to $\begin{pmatrix}56\\32\end{pmatrix}$

Find the value of x.

Question 6.

$$\boldsymbol{p} = \begin{pmatrix}8\\-28\end{pmatrix}$$

$\boldsymbol{p}$ is parallel to $\begin{pmatrix}-4\\y\end{pmatrix}$

Find the value of y.

Question 7.

$$\boldsymbol{a} = \begin{pmatrix}0\\4\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-3\\5\end{pmatrix}$$

If $3\boldsymbol{a} - 2\boldsymbol{b}$ is parallel to $\begin{pmatrix}x\\8\end{pmatrix}$, find the value of x.

Question 8.

$$\boldsymbol{a} = \begin{pmatrix}-2\\y\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-7\\-6\end{pmatrix}$$

If $4\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\begin{pmatrix}3\\-2\end{pmatrix}$, find the value of y.

Exam-Style Question

Given that:

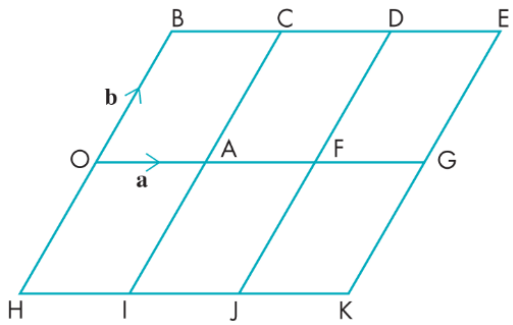
$$\boldsymbol{a} = \begin{pmatrix}2\\3\end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix}-4\\-1\end{pmatrix} \quad \boldsymbol{c} = \begin{pmatrix}5\\2\end{pmatrix}$$

a) Write $\boldsymbol{b} + \boldsymbol{c}$ as a column vector. (1)

b) Find a value of K such that $K\boldsymbol{a} + \boldsymbol{b}$ is parallel to $\boldsymbol{b} + \boldsymbol{c}$. (3)

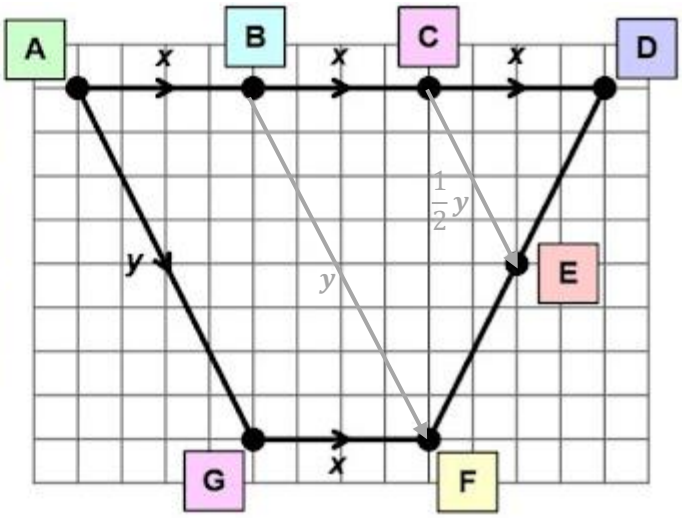
c) Give the resultant vector $K\boldsymbol{a} + \boldsymbol{b}$ as a column vector. (2)

1. Earlier on, we saw that adding and subtracting vectors can be seen as creating **a pathway**
2. **between two points**, with the resultant showing the 'direct' route between them.
3. We can use this, along with our knowledge of parallel vectors and properties of shapes to find
4. pathways between points in more complex geometrical figures.

Steps	*Example 1. I do...*	
1) Label the parallel vectors to those given in the diagram. 2) Follow a pathway between the given points. 3) Collect any like terms to simplify the final vector expression.	Write these vectors in terms of $\overrightarrow{OA} = \boldsymbol{a}$ and $\overrightarrow{OB} = \boldsymbol{b}$: *a)* $\overrightarrow{OF}$ b) $\overrightarrow{OH}$ c) $\overrightarrow{OC}$	
	We do...	***You do...***
	a) $\overrightarrow{OJ}$ b) $\overrightarrow{FH}$	*a)* $\overrightarrow{HD}$ b) $\overrightarrow{EJ}$

Vector Match: Match each of the vector expressions to a pathway below.

$2x - y$	$\frac{y}{2} + 2x$	y
$-2x$	$x - y$	$2x - \frac{1}{2}y$
$\frac{1}{2}y$	$-2x + y$	$3x - y$



$\overrightarrow{DF}$ =

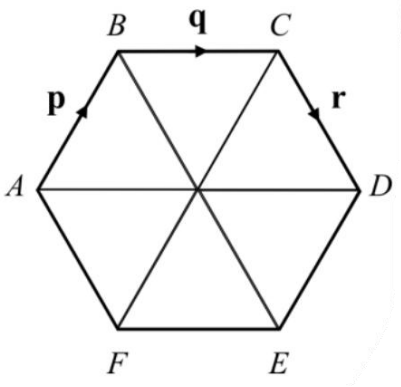
$\overrightarrow{BF}$ = $\overrightarrow{FC}$ = $\overrightarrow{GC}$ = $\overrightarrow{GD}$ =

$\overrightarrow{CE}$ = $\overrightarrow{GE}$ = $\overrightarrow{DB}$ = $\overrightarrow{AE}$ =

Question 1.

ABCDEF is a regular hexagon.

Find the vectors below in terms of $\boldsymbol{p}, \boldsymbol{q}$ and $\boldsymbol{r}$



a) $\overrightarrow{AB}$

b) $\overrightarrow{BA}$

c) $\overrightarrow{BC}$

d) $\overrightarrow{AF}$

e) $\overrightarrow{FE}$

f) $\overrightarrow{EF}$

g) $\overrightarrow{DE}$

h) $\overrightarrow{AC}$

i) $\overrightarrow{AD}$

j) $\overrightarrow{CE}$

k) $\overrightarrow{BE}$

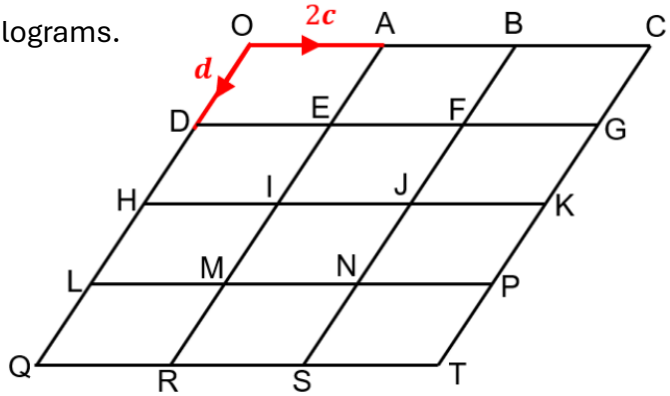
l) $\overrightarrow{DF}$

Question 2.

The diagram is made up of twelve congruent parallelograms.

$\overrightarrow{OA} = 2\boldsymbol{c}$ and $\overrightarrow{OD} = \boldsymbol{d}$.

Define the following vectors in terms of $\boldsymbol{c}$ and $\boldsymbol{d}$.



(a) $\overrightarrow{EG}$

(b) $\overrightarrow{BS}$

(c) $\overrightarrow{KJ}$

(d) $\overrightarrow{RE}$

(e) $\overrightarrow{OF}$

(f) $\overrightarrow{JT}$

(g) $\overrightarrow{MJ}$

(h) $\overrightarrow{QF}$

(i) $\overrightarrow{IL}$

(j) $\overrightarrow{HB}$

(k) $\overrightarrow{TE}$

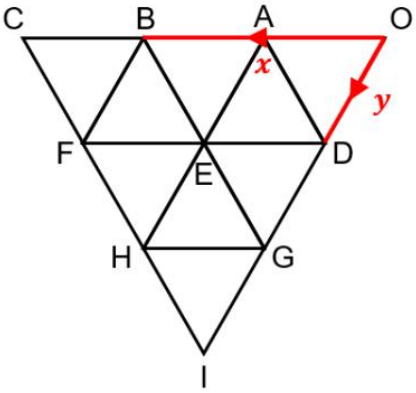
(l) $\overrightarrow{KD}$

Question 3.

The diagram is made up of nine congruent equilateral triangles.

$\overrightarrow{OB} = \boldsymbol{x}$ and $\overrightarrow{OD} = \boldsymbol{y}$.

Define the following vectors in terms of $\boldsymbol{x}$ and $\boldsymbol{y}$.



(a) $\overrightarrow{FD}$

(b) $\overrightarrow{HA}$

(c) $\overrightarrow{GH}$

(d) $\overrightarrow{OC}$

(e) $\overrightarrow{DB}$

(f) $\overrightarrow{OE}$

(g) $\overrightarrow{FO}$

(h) $\overrightarrow{CE}$

(i) $\overrightarrow{HI}$

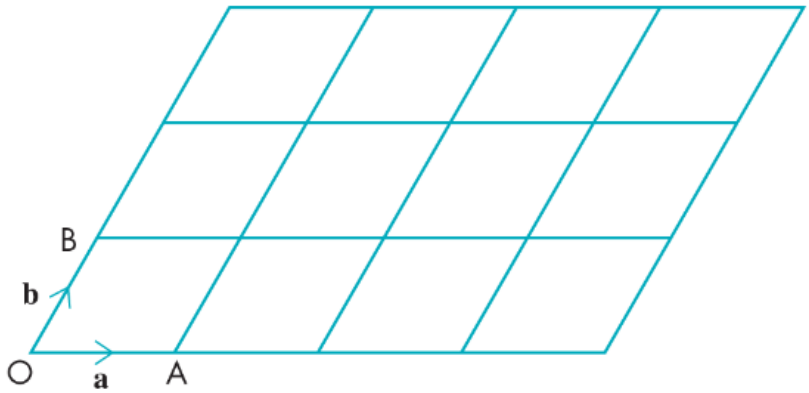
(j) $\overrightarrow{BG}$

(k) $\overrightarrow{HD}$

(l) $\overrightarrow{IF}$

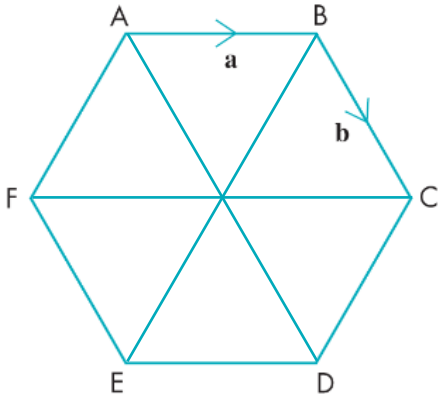
Question 4. Use the information below to mark points C-K on the grid.

a. $\overrightarrow{OC} = 2a + 3b$
b. $\overrightarrow{OD} = 2a + b$
c. $\overrightarrow{OE} = a + 2b$
d. $\overrightarrow{OF} = 3a + 3b$
e. $\overrightarrow{OG} = a + b$
f. $\overrightarrow{OH} = 2a + 2b$
g. $\overrightarrow{OI} = 3a + 2b$
h. $\overrightarrow{OJ} = 3b$
i. $\overrightarrow{OK} = 4a + b$



Question 5.

ABCDEF is a regular hexagon. $\overrightarrow{AB}$ is represented by the vector **a** and $\overrightarrow{BC}$ by the vector **b**.



a. By means of a diagram, or otherwise, show that $\overrightarrow{CD} = \mathbf{b} - \mathbf{a}$.

b. Express these vectors in terms of **a** and **b**.

i. $\overrightarrow{DE}$ ii. $\overrightarrow{EF}$ iii. $\overrightarrow{FA}$

c. Work out the resultant vector of adding:

$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} + \overrightarrow{EF} + \overrightarrow{FA}$

Explain your answer.

Q1.

$$\boldsymbol{a} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \text{ and } \boldsymbol{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

(a) Write down as a column vector

(i) $\boldsymbol{a} + \boldsymbol{b}$

.....

(1)

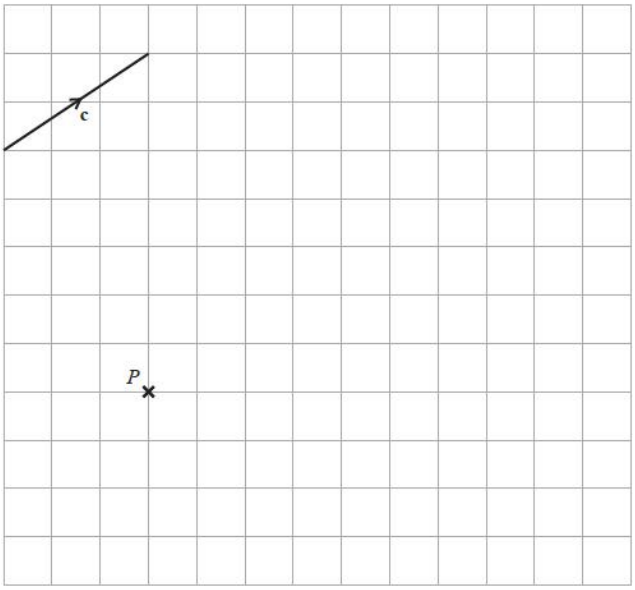
(ii) $2\boldsymbol{a} + 3\boldsymbol{b}$

.....

(2)

The vector $\boldsymbol{c}$ is drawn on the grid.

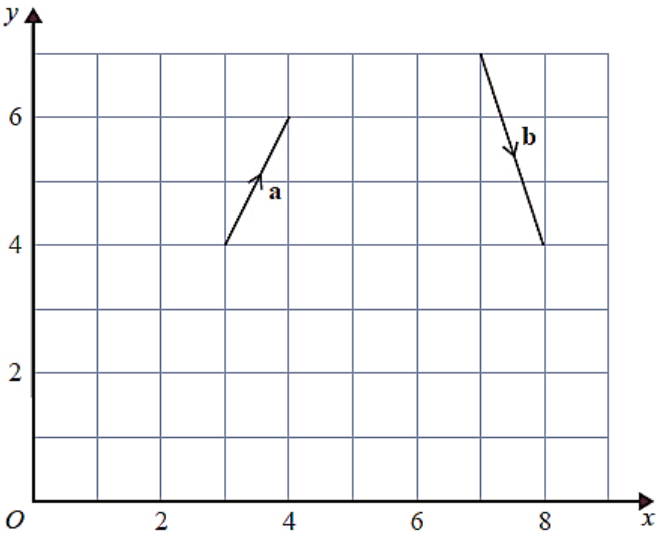
(b) From the point P, draw the vector $3\boldsymbol{c}$



(1)

(Total for question = 4 marks)

Q2. The vectors $\boldsymbol{a}$ and the vector $\boldsymbol{b}$ are shown on the grid.



a) On the grid, draw and label vector $-2\boldsymbol{a}$

(1 mark)

(b) Work out $\boldsymbol{a} + 2\boldsymbol{b}$ as a column vector.

(2 marks)

Q3. $\boldsymbol{a}$ and $\boldsymbol{b}$ are vectors such that:

$$\boldsymbol{a} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \text{ and } 3\boldsymbol{a} - 2\boldsymbol{b} = \begin{pmatrix} 8 \\ -17 \end{pmatrix}$$

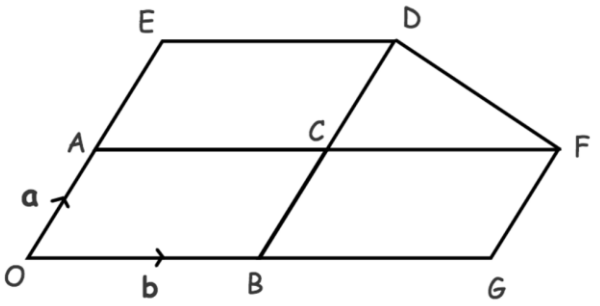
Find $\boldsymbol{b}$ as a column vector.

(3 marks)

Q4.
In the diagram OBDE and OAFG are parallelograms.

B is the midpoint of OG.
A is the midpoint of OE.

$\overrightarrow{OA} = \boldsymbol{a}$ and $\overrightarrow{OB} = \boldsymbol{b}$



a) Express in terms of a and b the following vectors, in their simplest form:

i. $\overrightarrow{OC}$

(1)

ii. $\overrightarrow{BA}$

(1)

iii. $\overrightarrow{DF}$

(1)

b) Show that $\overrightarrow{EG}$ and $\overrightarrow{DF}$ are parallel.

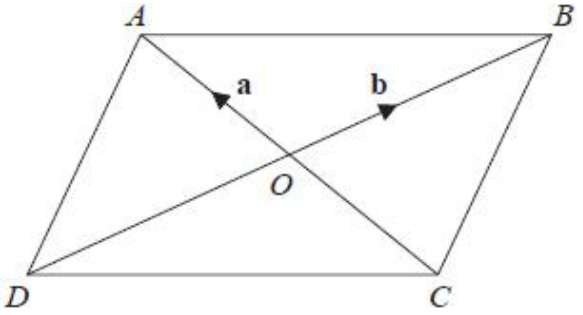
(2)

(Total marks for question is 5 marks)

Q5.
ABCD is a parallelogram.

The diagonals of the parallelogram intersect at *O*.

$\overrightarrow{OA} = \boldsymbol{a}$ and $\overrightarrow{OB} = \boldsymbol{b}$



a) Find, in terms of **b**, the vector $\overrightarrow{DB}$.

.....

(1)

(b) Find, in terms of **a** and **b**, the vector $\overrightarrow{AB}$.

.....

(1)

(c) Find, in terms of **a** and **b**, the vector $\overrightarrow{AD}$.

.....

(1)

(Total for question is 3 marks)